**STA304H1 Term Test 1**
**May 21, 2026 2:30 PM to 4:30 PM**

***Print your information in the following table:***

- Your STA304H1 term test 1 will be graded out of 35 points. This consists of 3 long answer questions Q1 (6 marks), Q2 (9 marks), Q3 (8 marks), and 12 multiple choice questions (1 mark each).
- Answer the following questions in the space provided on pages with a QR code. Any work that is not on a page with a QR code will not be graded.
- Please write clearly using a pencil with a HB or darker lead. Do NOT use erasable pens. Regular pens with a dark ink are acceptable.
- To earn the full credit, you must show all steps/work that demonstrate the logical steps and course concepts you are applying in your problem solving. Final answers are worth at most 1 point.
- You are NOT allowed to refer to any resources (including an electronic calculator).
- You must complete the test individually and are NOT allowed to discuss with others during the test.
- If the question asks for a numeric answer (*these are starred*), your final answer should be written as a single expression involving only numbers and standard arithmetic operations. That is, a calculator should be sufficient to compute the final value.

## Part A: Long Answer (23 marks)

Show all work for full credit.

**Q1. (6 marks)** Suppose the population ages are known:

$$y_1 = 48 \qquad y_2 = 23 \qquad y_3 = 39 \qquad y_4 = 31$$
$$y_5 = 41 \qquad y_6 = 36 \qquad y_7 = 29 \qquad y_8 = 50$$

The following sampling plan, for a sample of size $n = 4$, is proposed:

| Sample, $\mathcal{S}$ | $P(\mathcal{S})$ | $a$ | $8 \times a$ |
|---|---|---|---|
| $\mathcal{S}_1 = \{1,3,5,6\}$ | 1/8 | $(48+39+41+36)/4 = 41$ | 328 |
| $\mathcal{S}_2 = \{2,3,7,8\}$ | 1/4 | $(23+39+29+50)/4 = 32.25$ | 258 |
| $\mathcal{S}_3 = \{1,4,6,8\}$ | 1/8 | $(48+31+36+50)/4 = 41.25$ | 330 |
| $\mathcal{S}_4 = \{2,4,6,8\}$ | 3/8 | $(23+31+36+50)/4 = 35$ | 280 |
| $\mathcal{S}_5 = \{4,5,7,8\}$ | 1/8 | $(31+41+29+50)/4 = 37.75$ | 302 |

**a)** Do all units have the same probability of selection, $\pi_i$? If you answer no, provide a counterexample. If yes, state the common selection probability for each unit. [2 marks]

**b)** What is the exact sampling distribution of $\hat{t}_{\mathcal{S}} = 8\overline{y}_{\mathcal{S}}$, where $\overline{y}_{\mathcal{S}}$ denotes the sample mean? In context of the problem, what is $\hat{t}_{\mathcal{S}}$ an estimator for? [3 marks]

**c)** *Find the expectation of $\hat{t}_{\mathcal{S}}$, $\mathbb{E}[\hat{t}_{\mathcal{S}}]$.* [1 mark]

**Q2. (9 marks)** A simple random sample without replacement (SRS) of size 30 is taken from a population of size 100. The sample mean for income is $75,000 and the sample standard deviation is $10,000.

**a)** Show that the sample mean under SRS is an unbiased estimator of the population mean. Clearly define any notation used in your argument. [4 marks]

**b)** *Calculate a 95% confidence interval for the population mean income.* [2 marks]

| $z$ | $P(Z \geq z)$ |
|---|---|
| 1.64 | 5% |
| 1.96 | 2.5% |

Table 1: Standard Normal ($Z$) Table

| $P(t_\nu \geq t^*)$ | $\nu = 29$ | $\nu = 99$ |
|---|---|---|
| 5% | $t^* = 1.70$ | $t^* = 1.66$ |
| 2.5% | $t^* = 2.05$ | $t^* = 1.98$ |

Table 2: $t_\nu$-Distribution Table

**c)** *Let $\overline{y}_{\mathcal{U}}$ and $\overline{y}_{\mathcal{S}}$ denote the population and sample mean income, respectively. Solve for the required sample size $n$ that achieves the following desired precision:*

$$P(|\overline{y}_{\mathcal{S}} - \overline{y}_{\mathcal{U}}| \leq 1000) = 0.95$$

[3 marks]

**Q3.** **(8 marks)** A university wants to estimate the proportion of students who experience food insecurity, across a total population of 5,000 students. The population is stratified by housing type, and a total sample of size 100 is drawn **using proportional allocation**. The last column gives the **known** population standard deviation of the proportions within each stratum. *Note: Food insecurity means not having reliable access to enough affordable, nutritious food.*

| Housing type | $N_h$ | $n_h$ | $\hat{p}_{h\mathcal{S}}$ | $\frac{N_h}{N}\hat{p}_{h\mathcal{S}}$ | $\left(\frac{N_h-n_h}{N_h-1}\right)\left(\frac{N_h}{N}\right)^2\frac{p_h(1-p_h)}{n_h}$ | $\sqrt{p_h(1-p_h)}$ |
|---|---|---|---|---|---|---|
| On-campus residence | 2,500 | 50 | 0.10 | 0.05 | 0.00044 | 0.3 |
| Off-campus (with family) | 1,500 | 30 | 0.23 | 0.069 | 0.00047 | 0.4 |
| Off-campus (renting) | 700 | 14 | 0.43 | 0.0602 | 0.00033 | 0.49 |
| Temporary/unstable housing | 300 | 6 | 0.67 | 0.0402 | 0.00015 | 0.5 |
| Total (Sum) | 5,000 | 100 | 1.43 | 0.22 | 0.00139 | 1.69 |

Table 3: Stratified sample design by age group ($N = 5{,}000$, $n = 100$).

**a)** When solving for the variance of $\bar{y}_{\text{str}}$, what was $V(\bar{y}_{h\mathcal{S}})$ equal to? Justify your answer. [2 mark]

**b)** *The university administration claims that no more than 20% of students experience food insecurity and plans to use this figure to justify current funding for food support programs. Use Tables 1-3 to set up a hypothesis to test this claim at the 5% significance level. *Clearly state the null and alternative hypothesis, compute the observed test statistic, and the decision rule for rejecting or failing to reject the null hypothesis.* * [3 marks]

**c)** Under optimal (Neyman) allocation, how would observations be redistributed across strata compared to proportional allocation? Give an example in context of this problem. In simple terms, explain why optimal allocation is a beneficial allocation strategy to estimate the population proportion of students experiencing food insecurities.

[3 marks]

## Part B: Multiple Choice (12 marks)

To answer the following multiple choice questions, fill in the corresponding rectangle on the provided Crowdmark bubble sheet. **Make sure the MCQ number matches the question number on the bubble sheet.**

**MCQ1.** In a survey designed to study income, households are selected, and every person living in the selected household is interviewed. What is the observation unit?

(A) The list of all residential addresses

(B) A household listed in the sampling frame

(C) An individual person living in the household

(D) The variable recorded for each sampled unit (e.g. annual income)

(E) A household

**MCQ2.** In a survey study, which of the following **best describes** the sampled population?

(A) The group of units that meet eligibility criteria and could have been observed

(B) The group of units listed in the sampling frame that are a subset of the target population

(C) The group of units for which conclusions are intended

(D) The group of units listed in the sampling frame

(E) The specific subset of units that are randomly chosen to participate

**MCQ3.** Which of the following best describes what a sampling frame is?

(A) The complete list of observations a researcher wants to study

(B) The subset of the population selected for a study

(C) A list of sampling units from which a sample may be drawn

(D) A list of units that are actually observed and provide data in a study

(E) The numerical summary computed from a sample

**MCQ4.** The following questions are included in a public transit survey:

i. How many times per week do you use public transit?

ii. How satisfied are you with both the speed and cleanliness of public transit?

iii. Do you agree that public transit is poorly managed and needs more funding?

Which ONE of the following is true about the questions above?

(A) Question 3 is a leading question

(B) Question 1 is a double-barreled question

(C) Question 2 uses snowball sampling

(D) Question 1 is a leading question

(E) There are no double-barreled questions in the survey

**MCQ5.** A researcher stands at the entrance of a university library on a Tuesday afternoon and surveys the first 50 students who enter to estimate the average GPA of all students at the university. Which **source** of selection bias is **most clearly** present?

(A) Overcoverage

(B) Convenience sample

(C) Undercoverage

(D) Nonresponse

(E) Self-selection sample

**MCQ6.** A political polling company wants to estimate the proportion of Canadians who support a new federal housing policy. They conduct the poll by calling landline telephone numbers randomly selected from a national directory between 10am and 2pm on weekdays. Of the 500 numbers called, 180 people answer and complete the survey. Which of the following best describes the source(s) of bias in this study?

(A) Undercoverage, because people without landlines cannot be selected

(B) Nonresponse only, because the 64% who did not respond may differ systematically from those who did

(C) Self-selection only, because respondents chose to participate voluntarily

(D) Options (A) and (B)

(E) Options (B) and (C)

**MCQ7.** A researcher divides a city's population by neighbourhood and then randomly selects individuals within each neighbourhood. Which sampling design is this?

(A) Simple random sampling without replacement

(B) Systematic sampling

(C) Stratified random sampling

(D) Snowball sampling

(E) Cluster sampling

**MCQ8.** A health authority wants to survey patients at a large hospital but has no list of all patients. Instead, they randomly select 5 wards and survey everyone in those wards. This is an example of:

(A) Stratified random sampling

(B) Snowball sampling

(C) Systematic sampling

(D) Cluster sampling

(E) Simple random sampling without replacement

**MCQ9.** A simple random sample without replacement (SRS) of size $n = 3$ is drawn from a population of $N = 10$. What is the probability that any specific sample of 3 units is selected?

(A) $1/10$

(B) $3/10$

(C) $1/\binom{10}{3}$

(D) $1/10^3$

(E) $3/\binom{10}{3}$

**MCQ10.** Which sampling technique is most likely to produce biased results if the list from which units are selected has a periodic pattern related to the variable of interest?

(A) Simple random sampling without replacement

(B) Systematic sampling

(C) Stratified random sampling

(D) Cluster sampling

(E) Snowball sampling

**MCQ11.** Which ONE of the following statements is FALSE?

(A) The finite population correction substantially reduces the sampling error of an estimator under SRS when the sample size is much smaller than the population size

(B) The sampling weight represents the number of population units represented by each sampled unit in an estimator

(C) Under simple random sampling without replacement, sampling weights are always equal across all sampled units

(D) Under stratified sampling with proportional allocation, sampling weights are always equal across all sampled units

(E) When using stratified sampling, we sample in such a way that each stratum is represented in the sample

**MCQ12.** The stratified estimator of the population mean is defined as $\bar{y}_{\text{str}} = \sum_{h=1}^{H} \frac{N_h}{N} \bar{y}_{h\mathcal{S}}$. Which of the following correctly describes its bias?

(A) It is biased upward because larger strata receive more weight

(B) It is unbiased, since each $\bar{y}_{h\mathcal{S}}$ is an unbiased estimator of $\bar{y}_{hU}$ under SRS

(C) It is biased unless the sample sizes $n_h$ are equal across all strata

(D) It is unbiased only when proportional allocation is used

(E) It is biased unless all strata are the same size $N_h$